

A comparative study of Household Appliance Energy Forecasting.

Repository: <https://github.com/Priston11/appliance-energy-forecasting>

Course :Ms Data science

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1. Introduction

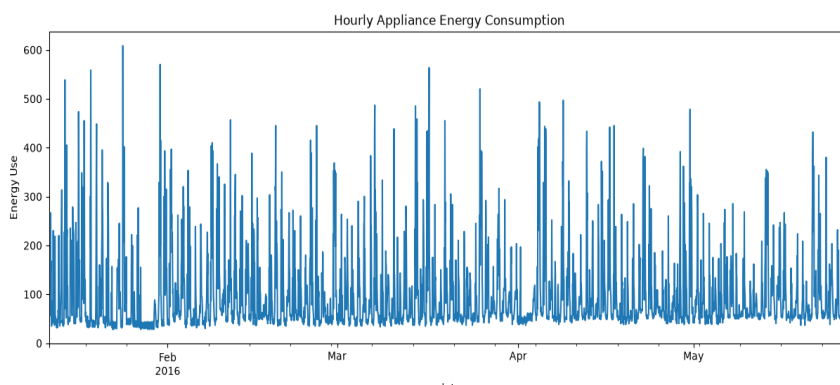
Short-term energy forecasting has become a key element of smart home energy management due to the integration of smart home technologies. The goal of this project is to predict daily energy demands of household appliances for 24 hours. This report investigates whether increasingly complicated models are worth their computational expense, and what model best fits use in the real world in smart homes, using a SARIMAX model, a machine learning model based on features (XGBoost), and a model foundation on time-series (Chronos) as benchmark models.

2. Data and Preprocessing

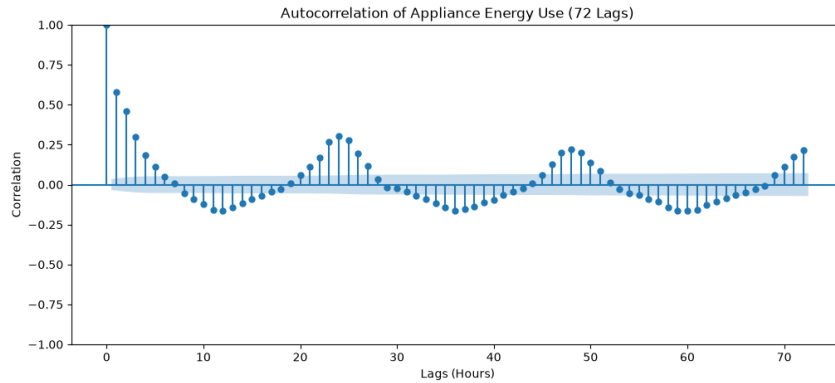
This Project is based on the appliances energy prediction dataset, which is used to measure the energy used by various appliances and applications along with other important indoor temperature, indoor humidity and outdoor weather data. All data were collected at 10 minute intervals. The data was resampled into hourly averages to make the modeling more manageable and pay attention to the shorter term daily trends. Appropriate handling of missing values, and a 24 h time scale period was defined. To avoid data leakage and to also test models with data which were not in the training set, the last 14 days were kept as a strict test set.

3. Exploratory Analysis

Preliminary analysis of the dependent variable (Appliances) showed clearly discernible cyclic patterns.



The time-series data exhibits noticeable peaks in energy consumption, following the common patterns of human activity (e.g., morning – evening activities). These patterns were further verified statistically by the use of autocorrelation function (ACF) plots..



The ACF plot showed very prominent spikes in the plot at lag intervals of 24 and 168 hours, indicating the presence of strong daily (24-hour) and weekly (168-hour) seasonality..

4. Forecasting Design

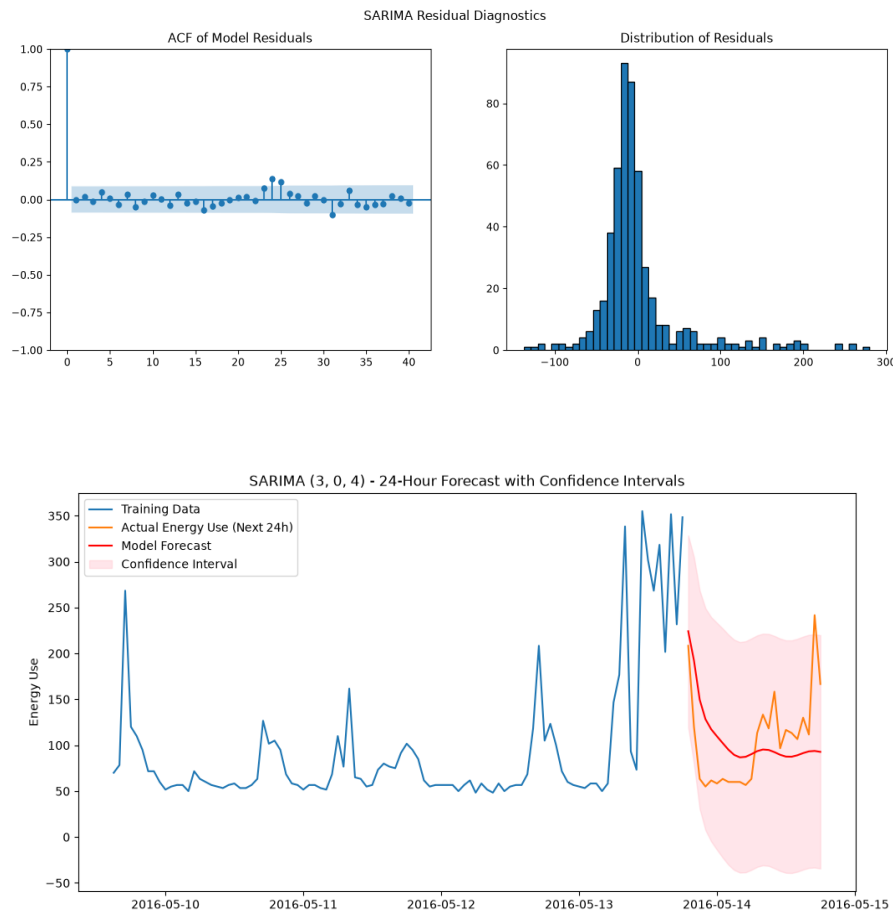
The forecasting problem is how much energy is going to be used by the appliances in the following 24 hours. The models were built with historical data and tested using the 14-day test period. Rolling features and lags were carefully shifted to guarantee a fair assessment and avoid data leakage: the models were only fed with the previous observations. The selection of the evaluation metrics was done using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Scaled Error (MASE), and Bias.

5. Benchmark Models

Some of the simple benchmark models were put to use to gain baseline measurement: The mean forecast is a flat average of the training data. Naive Forecast (aka extrapolation): Assuming the current value will continue in the future. Seasonal Naive (Daily & Weekly): Previous (yesterday) same hour (lag 24) or previous (last week) same hour (lag 168). Drift Forecast: Drawing a line or lines separating the first and last observation. The simplest model – the Weekly Seasonal Naive – was a good benchmark, as their behavior shows strong monthly resemblances (e.g. weekend patterns are similar to previous weekends).

6. SARIMAX Model

To capture both short-term autocorrelation and daily seasonality, a SARIMAX model was fitted. The model utilized an order of (1, 0, 1) and a seasonal order of (1, 1, 1, 24). Also, the variables were considered as exogenous variables: outdoor temperature (T_out) and outdoor humidity (RH_out).



The residuals also showed a fairly normal distribution of errors implying that the basic statistical structure was captured by the model..

7. Feature-Based Model

For complex and non-linear relationships, machine learning is employed to build an XGBoost model. Machine learning is then used to build an XGBoost model to handle complex, non-linear relationships. This model's feature table was built to contain the following:

- In addition, there is energy use for lagged appliances (leakage) which has been pushed off into the future. Rolling mean and standard deviations (rolling 24-hour periods).
- Hour-of-day, day-of-week indicators and weekend indicators. Weather covariates in and out of doors.

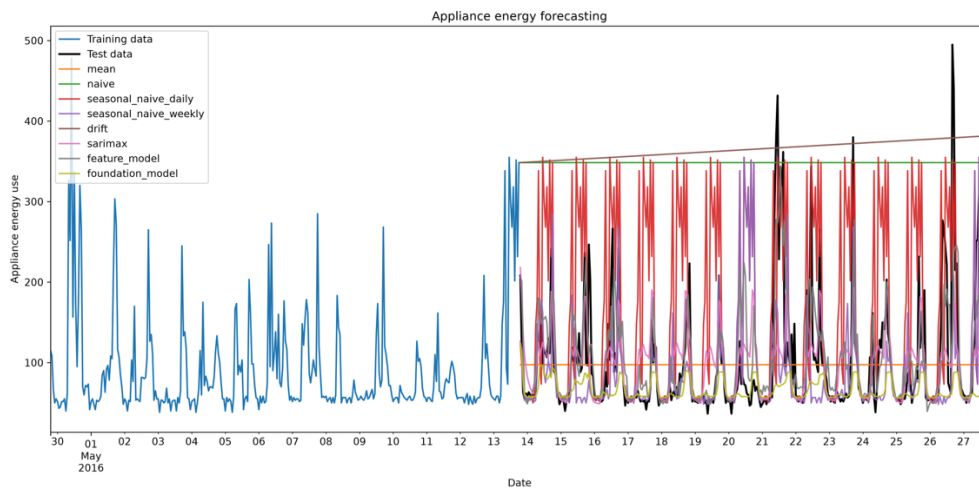
8. Foundation Model

The project additionally took a shot at Amazon's time-series base model Chronos. Like a language model predicting the next word, Chronos leans on "significant deep learning" to make predictions. It was used as a zero-shot forecasting model, i.e., it was required to predict the future energy consumption only from the past target sequence with no exogenous weather variables used.

9. Results and Error Analysis

Each model was tested on the last 14 days of data. The results of the evaluation metrics were as follows:

MAE	RMSE	MASE	Bias
34.23	55.77	0.64	6.47
36.63	64.16	0.68	-2.39
38.89	78.41	0.73	-34.24
42.63	79.29	0.80	-10.82
50.32	74.91	0.94	-3.11
86.96	129.23	1.63	64.01
250.00+	258.00+	4.60+	247.00+



The top overall performance was seen in the XGBoost (Feature Based) model which had the lowest MAE of 34.23 and a wonderful MASE of 0.64. The best result, by far, was obtained by SARIMAX (MAE 36.63) which outperformed the second best result (Weekly Seasonal Naive, MAE 42.63). This is a very impressive number for a zero-shot model designed without exogenous features, when compared to all of the benchmark models, and was achieved with the foundation model (Chronos).

10. Discussion and Limitations

One of the important factors for energy forecasting is that the amount and type of covariates at the forecast origin is available. Future time of the day/days of the week are quantities that are actually known beforehand.

Whereas the readings of future sensors in the house (such as room temperature) and actual outdoor weather parameters (such as wind speed, or precise humidity) are not precisely known 24 hours ahead. An additional important thing about this project was that XGBoost and SARIMAX models ran using the realised future values of the sensor and weather data of the test set which meant that these were considered to be conditional forecasts. These models would also depend on the weather forecasts which themselves have errors and might lead to a decrease in the model's accuracy in a real-life situation.

11. Conclusion

This research experiment compared various techniques of the household appliance energy prediction. The weekly seasonal naive model has been an extremely successful baseline, showing the predictable features of energy use by households.

The feature-based XGBoost model outperformed the Chronos foundation model in terms of accuracy, but the former demonstrates the ability of zero-shot AI forecasting. The Feature Based model (XGBoost) is suggested for developing a practical smart-home energy forecasting system.

It is very efficient and works well when incorporating complicated weather and behavioral covariates. But the system designers need to make sure it can be integrated with a trusted weather forecasting API, since its level of accuracy relies heavily on the availability of accurate weather forecasting.

12. Reference

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